Machine Learning with Applications 21 (2025) 100704



Contents lists available at ScienceDirect

Machine Learning with Applications

journal homepage: www.elsevier.com/locate/mlwa





A comprehensive overview of remaining useful life prediction: From traditional literature review to scientometric analysis

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ARTICLE INFO

Keywords:
Remaining useful life
Prognostics and health management
Model-based methods
Data-driven methods
Hybrid methods
CiteSpace
Scientometric analysis

ABSTRACT

The increasing complexity of industrial systems has heightened the need for precise Remaining Useful Life (RUL) predictions. This paper provides a comprehensive overview of RUL prediction methods, processes, datasets, and tools, alongside insights from scientometric analysis. We categorize RUL prediction methods into model-based, data-driven, and hybrid methods, detailing key models and their applications. Model-based methods utilize physical failure mechanisms, enhancing interpretability, but with limited adaptability for complex systems. Data-driven methods use machine learning to extensive datasets, boosting adaptability but often at the cost of explainability. Hybrid methods aim to strike a balance between these strengths, offering both accuracy and flexibility across various applications. In addition, we outline the RUL prediction process, from data acquisition to implementation, and introduce the primary datasets used in studies of engines, bearings, and batteries as well as commonly used programming frameworks and open-source libraries. We also performed a scientometric analysis of 3442 articles from the Web of Science database, examining research areas, prominent research groups, and emerging trends. Our findings offer a comprehensive overview of the RUL prediction field, outlining its current trends and highlighting potential directions for future research.

1. Introduction

The advent of Industry 4.0 has profoundly reshaped modern manufacturing by integrating IoT, big data analytics, and intelligent systems (Kagermann & Wahlster, 2022; Lasi et al., 2014). Among the technologies enabling this transformation, Prognostics and Health Management (PHM) plays a critical role by ensuring real-time monitoring, failure prevention, and predictive maintenance. Within PHM systems, Remaining Useful Life (RUL) prediction has emerged as a cornerstone for reducing unexpected downtimes and maintenance costs (Jardine et al., 2006; Lee et al., 2014).

Over the years, various methodologies have been developed for RUL prediction, which is generally categorized into three paradigms: model-based, data-driven, and hybrid methods (Zhang et al., 2023). Model-based methods rely on physical laws and state-space models such as Kalman Filters (Kalman, 1960), Extended Kalman Filters (EKF) (Gelb, 1974), and Particle Filters (PF) (Gordon et al., 1993) to simulate degradation behaviors (Daigle et al., 2012; Lei et al., 2013). Reviews by Cubillo et al. (2016), Li et al. (2024c), Wang et al. (2023) have provided in-depth summaries of such physics-based methods. In contrast, data-driven methods exploit statistical learning and machine learning to identify patterns in historical sensor data (Gebraeel et al., 2023). Algorithms like Support Vector Machines (SVM) (Widodo & Yang, 2007) and Long Short-Term Memory (LSTM) networks (Zheng et al., 2017) have shown great flexibility but often depend on large-scale, high-quality labeled datasets (Wang et al., 2008). Si et al. (2011) presented one of the earliest comprehensive reviews on statistical data-driven prognostics, laying a theoretical foundation for subsequent developments. To overcome the individual limitations of model-based and data-driven methods, hybrid models have been proposed. These methods aim to leverage the interpretability of models and the adaptability of learning-based models (Lei et al., 2016a; Zio, 2009). For example, Zio and Di Maio (Zio & Di Maio, 2010) integrated PFs with neural networks to enhance RUL estimation under uncertainty. A broader view of hybrid RUL models was presented in the works of Liao and Köttig (Liao & Köttig, 2014), Wang et al. (2023), and Li et al. (2024b), which explore applications across diverse industries and address recent advances in fusion strategies. Despite the technical richness of these methods, current reviews often focus narrowly on algorithmic taxonomies or application-specific evaluations. For instance, Lei et al. (2018) proposed a complete RUL prediction pipeline but did not address recent developments such as hybrid deep learning architectures. Pei et al. (2019) emphasized machine learning methods but omitted bibliometric perspectives. Recent reviews by Wang et al. (2023) and Li et al. (2024b) highlighted modern

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https://doi.org/10.1016/j.mlwa.2025.100704
Received 11 May 2025; Received in revised form 27 June 2025; Accepted 1 July 2025
Available online 18 July 2025
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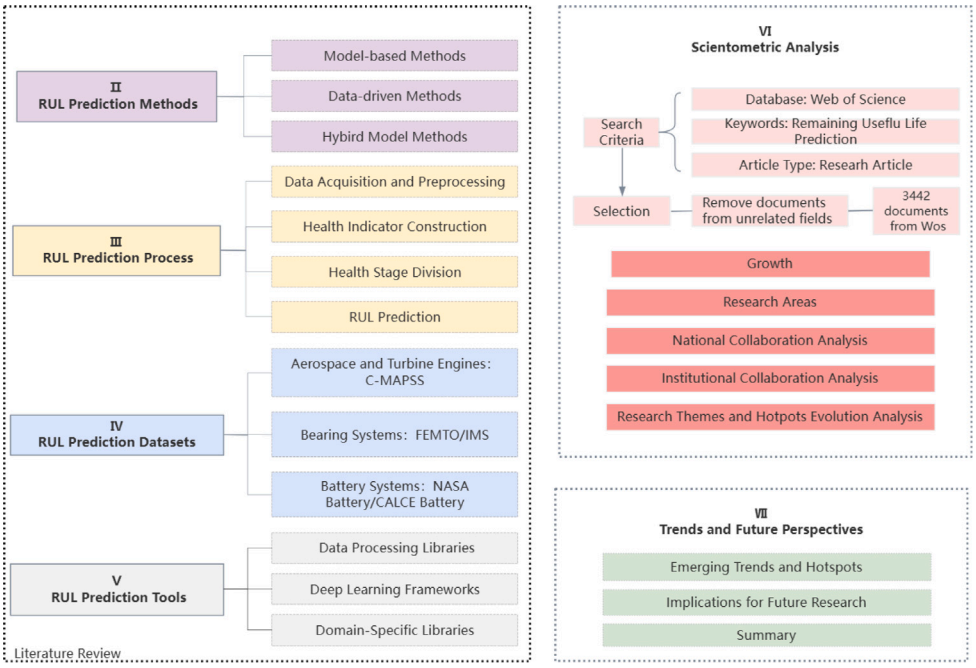


Fig. 1. Overall Structure of This Paper.

trends yet still lacked an integrated view of technical evolution and research dynamics.

To provide a comprehensive overview of the RUL prediction landscape, this paper first summarizes the three major prediction paradigms: model-based, data-driven, and hybrid methods. In addition, to support practical implementation, we then describe the typical RUL prediction process and introduce representative datasets (e.g., C-MAPSS, FEMTO-ST) along with commonly used implementation tools and open source libraries. Building upon this methodological foundation, we also conduct a scientometric analysis using CiteSpace. This includes keyword evolution, co-authorship networks, and thematic clustering based on bibliographic data from the Web of Science (WoS). By integrating methodological review with quantitative mapping, this dual-perspective survey offers both technical insight and a strategic understanding of emerging trends and research gaps.

The remainder of the paper is organized as follows. Section 2 reviews three major RUL prediction paradigms. Section 3 describes RUL prediction processes, while Section 4 and Section 5 summarize commonly used datasets and implementation tools. Section 6 presents the scientometric analysis. Finally, Section 7 discusses future research directions and concludes the study. An overview of the paper's structure is shown in Fig. 1.

2. RUL prediction methods

In this section, we categorize RUL prediction methods into three paradigms: model-based, data-driven, and hybrid methods based on existing work. We will detail some of these methods and their applications below.

2.1. Model-based methods

Model-based methods for RUL prediction leverage mathematical models to represent system degradation over time (Lee et al., 2014; Lei et al., 2016b). These include physical failure models that simulate degradation using laws like Archard's for wear and Paris' for crack growth. They also include state-space models that apply Kalman or PFs for real time health monitoring (Cubillo et al., 2016; Lee et al., 2014). Both methods require a deep understanding of system dynamics and are particularly effective for accurate, interpretable predictions essential for maintenance planning. Fig. 2 outlines some of the key subdivisions within model-based methods.

Model-based methods have been widely applied across various fields:

- **Aerospace engineering:** In this domain, fracture mechanics models (e.g., Paris' Law) predicted fatigue crack propagation by correlating crack growth rates with stress intensity factors, thereby supporting the maintenance of cyclically loaded components such as aircraft wings. In addition, stress–life models such as the SN curves and creep equations were instrumental in analyzing the behavior of the material under varying stress and high temperature conditions. For nonlinear systems, EKFs were frequently applied to enhance real-time health estimation. More details were provided in Cubillo et al. (2016), Lee et al. (2014), Liao and Köttig (2014).
- **Rotating machinery:** For rotating equipment, Archard's law was used to model wear in gears and bearings, while dynamic models predicted mechanical responses under variable loads. In addition, vibration-based methods supported early fault detection, and thermal models assessed overheating and lubrication failures. To handle complex degradation paths, PFs were employed, while Hidden Markov models (HMMs) tracked health states and predicted future conditions. More details were provided in Gordon et al. (1993), Widodo and Yang (2007).
- **Battery systems:** In electrochemical applications, models such as the Doyle–Fuller–Newman (DFN) model simulated internal reactions and aging mechanisms. Meanwhile, thermal and Solid Electrolyte Interphase (SEI) growth models addressed degradation caused by temperature fluctuations and capacity fade. Kalman filters were widely used for real time State of Charge (SOC) and State of Health (SOH) estimation, particularly when combined with optimization strategies. Further details were provided in Ansari et al. (2021), Qian and Liu (2021).
- **Wind turbines:** For wind energy systems, Miner's Rule predicted cumulative fatigue damage, while dynamic models evaluated structural responses to environmental loads. In parallel, creep and vibration-resonance models addressed long term deformation

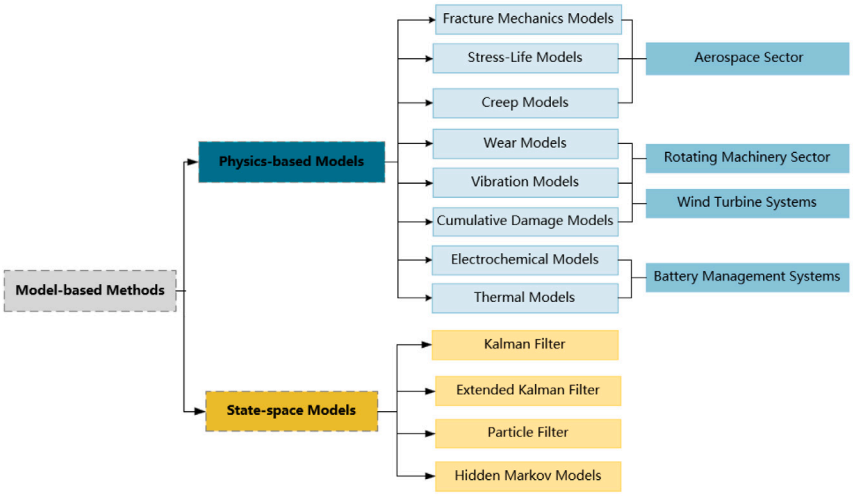


Fig. 2. Representative Model-based Methods for RUL Prediction.

and resonance-induced failures. PFs and HMMs were used to simulate degradation processes and monitor health transitions. More details were provided in Cubillo et al. (2016), Gordon et al. (1993).

2.2. Data-driven methods

With advancements in big data technology and computational power, data-driven methods have emerged as a dominant approach to RUL prediction, addressing the limitations of traditional physical models (Li et al., 2024c). These methods rely on analyzing historical and real-time operational data to extract features indicative of equipment health and lifespan. Data-driven methods include statistical models, such as Weibull distribution and Markov chains, as well as machine learning methods, which are further divided into shallow and deep learning approaches. Fig. 3 provides an overview of these methods. While statistical models face constraints from data availability and mathematical assumptions, machine learning methods like SVM (Widodo & Yang, 2007), Gaussian Process Regression (GPR) (Rasmussen, 2004), Artificial Neural Networks (ANN) (McCulloch & Pitts, 1943), Convolution Neural Network (CNN) (LeCun, 1989), and Recurrent Neural Network (RNN) (Rumelhart et al., 1986) have gained prominence for their adaptability and predictive performance.

2.2.1. Statistical models

Statistical models are foundational for RUL prediction, particularly for systems with observable degradation trajectories. These models can be categorized into four groups: continuous degradation process models (e.g., Wiener and Gamma process models), time-series models (e.g., ARMA models), state transition models (e.g., Markov chains), and statistical distribution-based models (e.g., Weibull models).

- **Wiener process models** are widely used to represent continuous and stochastic degradation over time. They are suitable for capturing non-linear wear, fatigue, and deformation patterns in mechanical components, and can handle random noise effectively (Li et al., 2018b). However, they are sensitive to non-Gaussian noise and often struggle with early stage predictions. Additionally, their applicability is limited when degradation is strictly monotonic. Extensions in Bayesian frameworks have been developed to improve uncertainty quantification and probabilistic inference. Further details can be found in Si et al. (2011), Zhang et al. (2018a).
- **Gamma process models** are effective for modeling monotonic, non-decreasing degradation behaviors. They are particularly suitable for long-term degradation scenarios such as lithium-ion batteries capacity fade, coating wear, and lubrication loss. Compared to Wiener process models, they offer better physical interpretability and computational efficiency, making them ideal for reliability assessment tasks. Further details can be found in Si et al. (2011), Van Noortwijk (2009).
- **AutoRegressive Moving Average models** are commonly applied to linear and stationary time-series degradation data. They perform well under clean, low-noise conditions and are mostly used for short-term prediction. However, they are less effective in nonstationary or noisy environments. ARMA models may also be used in multi-sensor settings, provided that data characteristics remain linear. Further details can be found in Pham and Yang (2010), Yuan et al. (2024).
- **Markov chain models** discretize system health into several states and are useful for systems with distinct degradation phases or health transitions. These models are particularly suited for applications like rotating machinery, where the system's health can be tracked discretely through several states. Further details can be found in Chiachío et al. (2020), Wang (2007).
- **Weibull distribution models** are a classic choice in reliability engineering for estimating failure times and system lifespans. They require a sufficient volume of failure data to achieve high prediction accuracy. However, their performance declines in early degradation stages or when training data is limited. Further details can be found in Ali et al. (2015), Weibull (1951).

To provide a concise comparison of these statistical models, Table 1 summarizes their key characteristics, strengths, limitations, and typical applications.

2.2.2. Support vector machine

SVM, introduced by Vapnik and Chervonenkis in the 1960s and further refined by Cortes and Vapnik in 1995 (Cortes & Vapnik, 1995), is widely used in supervised learning for pattern recognition and data analysis. Owing to its ability to maximize the margin between data classes, it has proven particularly effective in modeling complex systems under limited data conditions. In RUL prediction, SVM employs the following function for predictions (Tipping, 1999):

$$y(x) = \sum_{n=1}^{N} \omega_n K(x, x_n) + \varepsilon, \tag{1}$$

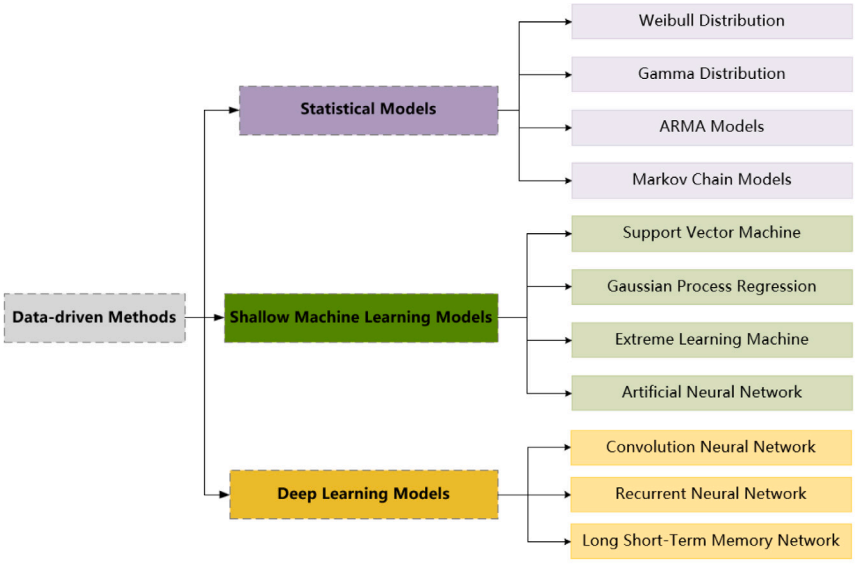


Fig. 3. Representative Data-driven Methods for RUL Prediction.

Table 1
Comparative analysis of statistical models for RUL prediction.

Model	Strengths	Limitations	Typical scenarios and representative works
Wiener Process Model	Models continuous degradation, handles Gaussian noise, suitable for nonlinear degradation, supports Bayesian inference for uncertainty quantification	Sensitive to non-Gaussian noise, poor early-stage predictions, inapplicable to strict monotonic degradation	Mechanical fatigue, material wear, turbine bearing degradation, (Si et al., 2011; Zhang et al., 2018a)
Gamma Process Model	Suited for monotonic, non-decreasing degradation, physically interpretable, analytical tractability for additive degradation	Cannot handle degradation reversal or non-monotonic behavior	Battery aging, lithium-ion batteries capacity degradation, (Si et al., 2011; Zhang et al., 2018a)
ARMA Model	Suitable for linear time-series data, models dependencies in degradation over time, simple and efficient implementation	Struggles with non-stationary data, poor performance in high-noise conditions, limited in multi-sensor fusion due to linear assumptions	Vibration data for motors, electrical equipment, or simple time-series degradation data, single-sensor temperature monitoring, (Yuan et al., 2024)
Markov Chain Model	Models discrete degradation states and transitions effectively, useful for multi-level health status prediction	Not suitable for continuous degradation, requires predefined states and transitions	Systems with discrete health states, such as rotating machinery, hydraulic system pumps, (Wang, 2007)
Weibull Distribution Model	Widely used and understood, suitable for modeling life distributions, commonly used in reliability engineering	Requires abundant failure data, less effective for early-stage predictions or sparse data	Lifetime prediction of large-scale machinery, aerospace components, industrial systems, (Weibull, 1951)

where ω_n represents the model weights that connect the feature space to the output, $K(\cdot)$ is the kernel function, and ε is the independent noise term.

Over the past three decades, numerous studies have explored the application of SVM and its extended variants in RUL prediction, demonstrating their effectiveness across various industrial systems. For instance, Liu et al. (2005) pioneered the use of SVM in RUL prediction, highlighting its advantages under limited sample conditions. Subsequently, several variants were introduced, such as Least Squares SVM (LS-SVM) (Dong & Luo, 2013), Multivariate SVM (MSVM) (Chen et al., 2013), and Support Vector Regression (SVR) (Benkedjouh et al., 2015; Wang et al., 2015), which have been widely applied to systems including lithium-ion batteries, bearings, and aero engines.

To enhance prediction accuracy, researchers have also proposed task-specific adaptations. For example, Patil et al. (2015) developed a multi-stage SVM model that segments battery life into distinct phases. In addition, hybrid methods combining SVM with other algorithms have emerged as a promising direction. Such efforts include Zhe et al. (2019), who integrated Kernel Principal Component Analysis (KPCA) with LS-SVM for bearing RUL prediction, and Zou et al. (2021), who combined ANN with SVM to improve performance.

However, despite these advancements, SVM-based models often require careful parameter tuning and are limited in their ability to capture long-term temporal dependencies—an essential aspect of degradation modeling. These limitations have spurred interest in alternative approaches better suited for sequential data and complex degradation dynamics. Further details can be found in Huang et al. (2015), Yan et al. (2020).

2.2.3. Gaussian process regression

GPR is a non-parametric Bayesian regression method that predicts outcomes while simultaneously quantifying uncertainty (Kang, 2019; Rasmussen, 2004). The predictive framework of GPR consists of the following sequential components:

Step 1: Latent Function Definition. GPR defines a latent function $f(x)$ over the input space as:

$$f(x) \sim N\left(\mu(x), k(x, x')\right), \tag{2}$$

where $\mu(x)$ is the mean function and $k(x, x')$ is the covariance (kernel) function.

Step 2: Kernel Specification. A squared-exponential kernel is commonly employed to model input correlations:

$$k_{SE}(x,x') = \sigma_f^2 \exp\left[-\frac{1}{2l^2}(x-x')^2\right], \tag{3}$$

with signal variance σ_f^2 and length-scale l.

Step 3: Observation Model Formulation. The observation model is expressed as:

$$y = f(x) + \epsilon, \tag{4}$$

where $\epsilon \sim N(0,\sigma^2)$ denotes i.i.d. Gaussian noise. The prior over any finite set of outputs is therefore

$$y \sim N\big(0,\ K(X,X) + \sigma_n^2 I_n\big). \tag{5}$$

Step 4: Joint Prior Construction. The joint prior of training outputs y and test outputs f_* can be written as:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim N\Big(0,\ \begin{bmatrix} K(X,X)+\sigma_n^2 I_n & K(X,x_*) \\ K(x_*,X) & k(x_*,x_*) \end{bmatrix}\Big).$$

Step 5: Posterior Inference. The posterior for the prediction f_* is:

$$f_* \mid X, y, x_* \sim N\big(\bar{f}_*, \mathrm{cov}(f_*)\big),$$

with

$$\bar{f}_* = K(x_*,X)\left(K(X,X)+\sigma_n^2 I_n\right)^{-1} y, \tag{6}$$

and

$$\mathrm{cov}(f_*) = k(x_*,x_*) - K(x_*,X)\left[K(X,X)+\sigma_n^2 I_n\right]^{-1} K(X,x_*), \tag{7}$$

respectively.

GPR has been extensively applied to RUL prediction, and many studies focus on kernel design. For example, Liu et al. (2019a) embedded the Arrhenius law into a polynomial kernel to reflect lithium-ion batteries ageing. Similarly, Hong et al. (2015) proposed composite kernels for bearing RUL to capture multi-level degradation. In another study, Feng and Shi (2021) combined grey correlation analysis with polynomial regression to refine GPR features. Wilson and Adams (2013) introduced the spectral-mixture kernel, thereby improving pattern discovery and long-range extrapolation. To enhance robustness, Aye and Heyns (2017) employed an ensemble of simple mean and covariance functions for rolling bearings, while Liu and Chen (2019) used indirect health indicators and multiple GPR models to address capacity-measurement issues in operating batteries. Yet, GPR still faces challenges in scalability, kernel engineering, and temporal sequence modeling, which has prompted exploration of more adaptive, data-driven alternatives. Further details can be found in Guo et al. (2023).

2.2.4. Artificial neural network

ANN is well-suited for complex nonlinear problems and can achieve high accuracy in RUL prediction. It comprises layers of interconnected neurons, where information flows from the input layer through hidden layers to the output layer (McCulloch & Pitts, 1943). The mathematical representation of a neuron is:

$$y = f\left(\sum_{i=1}^{n}(w_i x_i) + b\right). \tag{8}$$

Early studies demonstrated the potential of ANN in RUL prediction. For example, Parthiban et al. (2007) used ANN to model lithium-ion batteries performance under varying conditions. Wu et al. (2016) applied ANN with importance and systematic sampling to enhance feature extraction for model training. Tavakoli et al. (2019) combined ANN with adaptive neuro-fuzzy inference systems to predict equipment RUL, while Wu et al. (2018) employed a systematic ANN-based approach for battery RUL estimation. To improve prediction accuracy, Li et al. (2016) introduced a PF method based on an ensemble ANN, validating its feasibility in lithium-ion batteries simulations. As neural networks advanced, diverse ANN architectures have been applied to RUL prediction, reflecting their adaptability and effectiveness. However, standard ANN architectures may struggle with temporal dependencies in sequential data, leading to increased interest in recurrent and attention-based models. Further details can be found in Li et al. (2023a).

2.2.5. Convolution neural network

CNN, as a representative deep learning model, is particularly effective in processing multi-dimensional sensor data (LeCun, 1989). Typically, time-series data are transformed into a two-dimensional "time-feature" matrix, enabling CNNs to automatically extract hierarchical features through stacked convolution and pooling layers.

The convolution operation is defined as:

$$y^{l(i,j)} = K_i^l * x^{l(r^j)} = \sum_{j'=0}^{W-1} K_i^{l(j')} x^{l(j+j')}, \tag{9}$$

where W denotes the width of the convolution window, $K_i^{l(j')}$ represents the j'-th weight of the ith kernel in the lth layer, and $x^{l(r^j)}$ is the receptive field at position j.

To enhance nonlinearity and sparsity, CNNs typically apply a rectified linear unit activation function to the convolution outputs, helping mitigate overfitting:

$$a^{l(i,j)} = f\left(y^{l(i,j)}\right) = \max\left\{0, y^{l(i,j)}\right\}, \tag{10}$$

where $a^{l(i,j)}$ is the post-activation output.

Subsequently, a pooling layer is introduced to perform downsampling. Among these, max pooling is widely used, formulated as:

$$p^{l(i,j)} = \max_{(j-1)V+1 \le t \le jV}\left\{a^{l(i,t)}\right\}, \tag{11}$$

where V indicates the pooling width, $p^{l(i,j)}$ is the pooled output, and $a^{l(i,t)}$ denotes the input activation.

Thanks to their powerful feature extraction capabilities, CNNs have been increasingly adopted for RUL prediction tasks. For example, Li et al. (2018a) first introduced CNNs into RUL estimation, demonstrating their effectiveness in handling temporal sensor signals. Later, Li and Yang (2020) developed a hyperparameter-optimized CNN tailored for lithium-ion batteries RUL prediction. Similarly, Yang et al. (2019) proposed a dual-CNN framework to enhance spatial and temporal feature fusion. To improve perceptual capacity, Hong et al. (2020) introduced dilated CNNs, enabling better degradation trend identification within fewer operational cycles. Deep CNNs (DCNNs) also demonstrated superior performance in RUL prediction. For instance, Ren et al. (2018) employed a spectral principal energy vector in DCNNs to process bearing vibration signals. Zhou et al. (2020) applied temporal CNNs to battery SOH and RUL prediction, while Cheng et al. (2021) improved generalization with a transferable CNN framework under cross-domain scenarios. In addition, Yang and Ji (2021) proposed a CNN grid model for processing multidimensional features from both time and time–frequency domains. Ma et al. (2020) enhanced RUL accuracy for aviation engines with an improved CNN. Sun et al. (2021) introduced a sorting feature method using Temporal Convolutional Networks (TCNs) to further boost prediction performance in rotating machinery. However, despite CNNs' strengths in spatial feature extraction, their limitations in modeling long-range temporal dependencies remain evident. This limitation has driven the development of sequence-oriented architectures that better capture temporal dynamics. Further details can be found in Li and Yang (2021), Ren et al. (2020).

2.2.6. Recurrent neural network

RNN is a specialized neural network designed for sequential data, with a recurrent structure that captures temporal dependencies (Rumelhart et al., 1986). Unlike feedforward networks, RNN relies on hidden states, where the current state depends on both the current input

and the previous state. The hidden state and output are calculated as follows:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b_h), \tag{12}$$

and

$$y_t = W_y h_t + b_y, \tag{13}$$

respectively, where W_h, W_x, and W_y are weight matrices, b_h and b_y are biases, and tanh is the activation function.

RNNs have been widely adopted for RUL prediction due to their ability to process time-dependent sensor signals. For example, Liu et al. (2010) developed an adaptive RNN model for RUL prediction, improving reliability and accuracy in practical scenarios. Chaoui and Ibe-Ekeocha (2017) introduced a dynamically driven RNN for SOC and SOH estimation in lithium-ion batteries, while Heimes (2008) employed RNN with backpropagation through time and EKF algorithms for RUL prediction. In terms of architectural advancements, the memory-optimized RNN proposed by Liu et al. (2014) for battery dynamics, and the competitive learning mechanism developed by Malhi et al. (2011) for defect propagation prediction, have further propelled the application of RNNs in RUL prediction.

However, standard RNNs often struggle with learning long-range dependencies due to the vanishing or exploding gradient problem. To tackle this issue, the LSTM network was introduced as an advanced variant. By incorporating a gating mechanism — including the input gate, forget gate, and output gate — LSTM networks are capable of capturing both short-term fluctuations and long-term degradation patterns (Hochreiter & Schmidhuber, 1997). Specifically, the 'forget gate' determines which parts of the previous cell state C_{t-1} should be discarded, the 'input gate' controls what new information is added to the memory cell, and the 'output gate' regulates the output of the current hidden state. The calculation formulas for each time step are as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \tag{14}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \tag{15}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \tag{16}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \tag{17}$$

where f_t is the forget gate, i_t is the input gate, $\tilde{C}_t$ is the candidate cell state, and o_t is the output gate. Depending on the value of the forget gate and the candidate cell state, the value of the cell state is calculated as:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t. \tag{18}$$

Then, the current hidden state h_t is determined by the value of the output gate and the tanh function of the cell state C_t:

$$h_t = o_t \cdot \tanh(C_t). \tag{19}$$

LSTM networks have been widely applied in RUL prediction due to their capability to capture long-term temporal dependencies (Zheng et al., 2017). For example, Yuan et al. (2016) applied LSTM to aircraft engine fault diagnosis and RUL prediction, demonstrating accurate predictions of engine lifespan. Similarly, Guo et al. (2017) incorporated time-frequency features into LSTM for bearing degradation prediction, overcoming gradient issues.

To further enhance prediction performance, researchers have explored various LSTM variants. One such example is the bidirectional LSTM (Bi-LSTM), studied by Zhang et al. (2018c), which leverages forward and backward learning to improve RUL predictions. Meanwhile, continuous efforts in architecture innovation have led to the emergence of advanced LSTM-based models. Chinomona et al. (2020) integrated statistical sampling methods into LSTM to handle incomplete data environments, while Wu et al. (2021) introduced degradation-aware LSTM autoencoders to better capture equipment degradation patterns. In the context of fuel cells, Ma et al. (2018) developed a Grid LSTM to optimize prediction accuracy, and Chang et al. (2022) proposed an multi-layer grid search optimized LSTM algorithm that mitigates the impact of initial prediction time on RUL prediction accuracy. More recently, there has been a growing trend of integrating LSTM with other models and feature engineering techniques to further improve the robustness and accuracy of RUL predictions. Further details can be found in Huang et al. (2025), Ma and Mao (2020).

2.3. Hybrid model methods

In RUL prediction, model-based and data-driven methods each face distinct challenges. Model-based methods, grounded in physical principles, often lack adaptability to complex or dynamic systems, while data-driven methods rely heavily on data quality and quantity, with limited interpretability. Hybrid models address these limitations by integrating complementary strengths, thereby improving precision and robustness in complex systems where standalone methodologies often falter. These models provide a balanced and adaptable framework for predictive maintenance, drawing increasing scholarly attention. Fig. 4 illustrates selected examples of hybrid models developed for RUL prediction, highlighting representative approaches rather than an exhaustive list.

2.3.1. Model and data-driven hybrid models

Hybrid models combine model-based and data-driven methods to leverage their respective strengths. Model-driven methods provide theoretical guidance, enhancing interpretability, while data-driven methods improve adaptability and flexibility in dynamic environments, addressing the predictive needs of complex systems.

A variety of studies have demonstrated the effectiveness of hybrid strategies. For example, Zemouri and Gouriveau (2010) combined the recurrent radial basis function network with the autoregressive (AR) model, achieving more precise predictions by merging the advantages of both approaches. Yang et al. (2016) integrated a double exponential empirical model with a Relevance Vector Machine (RVM) for lithium-ion batteries RUL prediction, demonstrating strong performance. Similarly, Zhao et al. (2019) proposed a hybrid model based on Gray Models (GM) and RVM, alternating between the two for multistep SOH predictions, achieving high accuracy. Hervé de Beaulieu et al. (2024) explored physics-informed data augmentation to integrate system knowledge into data-driven predictions, effectively reducing the need for large labeled datasets. PFs are often used as bridging techniques in hybrid models. Li and Xu (2015) combined GPR with PFs for robust predictions under uncertainty, while Wu et al. (2019) used neural networks with bat-based PFs for RUL prediction. Zhang et al. (2021) enhanced accuracy by integrating an adaptive Levy flight optimization PF with LSTM networks, outperforming traditional single-method approaches. A prominent emerging paradigm is Physics-Informed Neural Networks (PINN), which explicitly incorporate physical constraints into the loss function, enforcing the network to comply with physical laws. This approach not only improves prediction accuracy but also reduces data dependency and enhances model interpretability. Liao et al. (2023) combined PINNs with self-attention to enhance prediction accuracy and interpretability. Their approach outperformed conventional data-driven models on the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset, achieving better generalization with lower complexity. E et al. (2025) proposed a PINN framework integrating RUL prediction with degradation modeling featuring three components, enhancing interpretability and generalizability under limited data. Further details can be found in Li et al. (2024c), Wang et al. (2023).

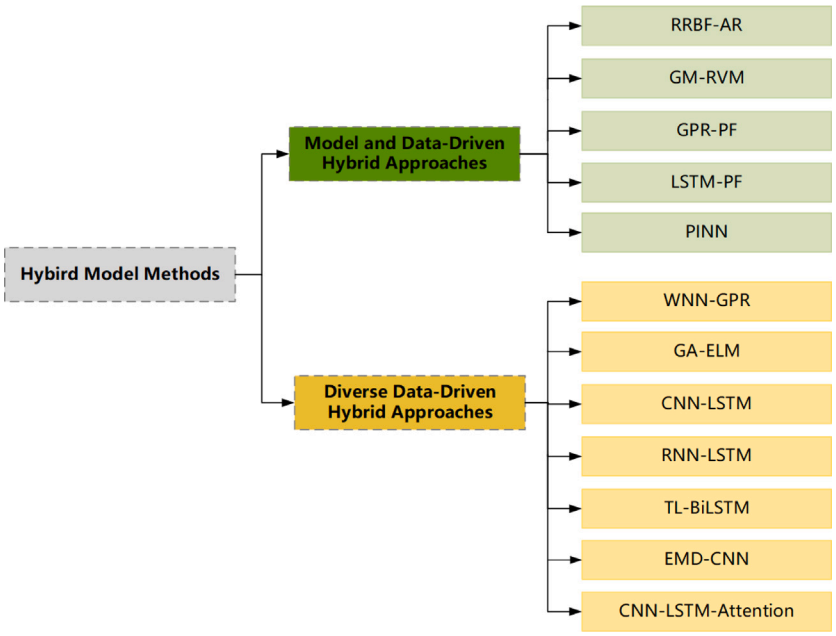


Fig. 4. Representative Hybrid Models for RUL Prediction.

2.3.2. *Diverse data-driven hybrid methods*

Combining various data-driven methods leverages information from multiple dimensions, enhancing the accuracy and robustness of RUL predictions by capturing complex system behaviors (Liu et al., 2024a).

A great deal of hybrid frameworks have been proposed to exploit complementary strengths across models. For example, Hong et al. (2014) proposed an adaptive prediction method by integrating Wavelet Neural Networks (WMNN) with GPR. Similarly, Zhang et al. (2018b) utilized Bi-LSTM and transfer learning to enhance prediction performance by transferring knowledge between domains. Dong et al. (2017) developed an RNN-LSTM hybrid model for RUL prediction. Li et al. (2019) integrated LSTM with Elman neural networks to simultaneously capture long-term dependencies and short-term signal dynamics. Wang et al. (2020) optimized LS-SVM using an improved bird swarm algorithm for better RUL predictions, and Chen et al. (2020) combined Genetic Algorithms (GA) with Extreme Learning Machines (ELM) to predict voltage degradation in fuel cells. Ren et al. (2020) enhanced accuracy by combining autoencoders with CNN-LSTM, while Li and Yang (Li & Yang, 2021) proposed a CNN-LSTM model for lithium-ion batteries RUL prediction, effectively addressing nonlinear degradation. Huang et al. (2025) proposed a CNN-LSTM model integrated with a temporal pattern attention mechanism to enhance the state of health estimation accuracy for lithium-ion batteries, effectively improving SOH prediction. Recent advances in signal decomposition-enhanced deep learning have addressed partial discharge data challenges. For instance, Liu et al. (2024a) developed a variational mode decomposition Variational Mode Decomposition (VMD)-optimized CNN-BiGRU framework, which eliminates the need for complete charge–discharge cycles. Neural network integration is a growing focus in hybrid methods. While hybrid methods demonstrate superior accuracy compared to single methods, they often involve increased complexity in implementation and maintenance. Further details can be found in Li et al. (2024c), Liu et al. (2024a).

2.4. *Comparative summary of RUL prediction methods*

To facilitate a holistic understanding of the three primary RUL prediction paradigms — model-based, data-driven, and hybrid methods — this section provides a structured comparison across key evaluation dimensions. The comparison helps identify the strengths, limitations, and applicability of each method category, supporting the selection of appropriate techniques in practical scenarios.

As shown in Table 2, model-based methods are preferable when physical degradation mechanisms are well understood. In contrast, data-driven methods offer scalability and strong performance in scenarios with abundant data. Hybrid methods offer a promising direction by integrating the strengths of both paradigms, although they often incur higher computational and implementation costs. Thus, the choice of method depends on application requirements, data availability, and system complexity.

Recent advances build on the strengths summarized above and show where each paradigm is now heading. Transformer-style networks and graph neural networks like GraphRUL push beyond traditional CNN/LSTM baselines by learning long-range temporal features and component-to-component dependencies (Liao et al., 2023; Wang et al., 2025b). These examples illustrate a general shift from simple model stacking toward tightly integrated solutions that incorporate physical knowledge and minimize data requirements, pointing to the rich research space between three paradigms outlined in Table 2.

3. RUL prediction process

RUL prediction is a critical component of machinery health prognostics, involving a structured four-stage process (Lei et al., 2016a): data acquisition, health indicator construction, health stage division, and RUL prediction. The detailed descriptions are as follows:

Step 1 involves collecting sensor data from critical components to monitor equipment condition and capture early signs of degradation continuously.

Step 2 applies signal processing and machine learning techniques to extract features and construct health indicators that represent machine health.

Step 3 uses these health indicators to divide the system's condition into distinct stages, each corresponding to a different phase of degradation progression.

Step 4 predicts the RUL by analyzing the degradation trends observed in the later stages, comparing them against a predefined failure threshold to estimate the time to failure (Lei et al., 2018).

This systematic process supports informed maintenance decisions and enhances operational efficiency, as shown in Fig. 5.

Table 2
Comparative analysis of RUL prediction methods.

Method type	Strengths	Limitations	Typical scenarios	Representative works
Model-based Method	High interpretability; grounded in physical principles; effective for systems with known degradation laws	Requires expert knowledge of system dynamics; limited adaptability to changing environments or complex systems	Aerospace components, rotating machinery, systems with clear physical models	Lee et al. (2014), Cubillo et al. (2016)
Data-driven Method	Flexible and scalable; learns from data without physical assumptions; performs well with rich sensor inputs	Requires large labeled datasets; performance may degrade in sparse/noisy data; lacks physical interpretability	Lithium-ion batteries, bearings, industrial equipment with abundant run-to-failure data	Liu and Chen (2019), Ferreira and Gonçalves (2022), Pei et al. (2019)
Hybrid Method	Combines the interpretability of physics-based models with the flexibility of data-driven learning; robust in complex and uncertain conditions	Higher implementation complexity; computationally demanding; requires careful model fusion and tuning	Critical systems like aero-engines, fuel cells, or dynamic systems with non-stationary conditions	Liao and Köttig (2014), Zhang et al. (2021), Wang et al. (2023)

3.1. Data acquisition and preprocessing

Data acquisition is the foundational step in RUL prediction, involving the collection of monitoring data from sensors strategically placed on equipment (Wang et al., 2019). Common sensors, such as vibration, temperature, voltage, and pressure sensors, are selected based on equipment failure modes. For instance, vibration sensors detect early wear in rotating machinery, while temperature sensors monitor thermal stresses in engines and electrical systems (Lei et al., 2018). Advanced signal processing techniques, like wavelet transforms, are often employed to enhance feature extraction. Since it is hard to collect real data all the way until a machine fails, researchers often use simulated scenarios or lab experiments instead. While useful, these approaches may not fully capture real-world conditions. As a result, simulated datasets are widely used in RUL research, with subsequent sections of this paper discussing key datasets across various fields.

3.2. Health indicator construction

After data acquisition, feature extraction methods such as time domain analysis (e.g. Root Mean Square (RMS), kurtosis) and frequency domain analysis (e.g. spectral characteristics) are used to develop Health Indicators (HI) that quantify system health and reveal degradation processes (Zuo et al., 2021). Effective HIs simplify the modeling and enhance prediction accuracy by reflecting the deterioration of machinery with properties such as monotonicity, robustness, and trendability (Lei et al., 2018).

To further categorize HIs, they are commonly divided into Physics-based Health Indicators (PHIs) and Virtual Health Indicators (VHIs) (Hu et al., 2012). The former are derived from physical mechanisms of failure, use statistical or signal processing methods, such as RMS or variance, to monitor wear in rotating machinery (Večeř et al., 2005; Zhu et al., 2014). The latter, in contrast, integrate multiple PHIs or sensor signals to provide a more comprehensive representation of system degradation, which is particularly useful for systems exhibiting multiple failure modes (Lei et al., 2018). Although VHIs may lack direct physical interpretability, they enable holistic health assessment by synthesizing information across different domains. To construct VHIs, dimensionality reduction methods such as PCA and self-organizing maps are widely employed, as they help reduce noise and redundancy while preserving essential degradation patterns (Lei et al., 2018). Moreover, advanced machine learning models — such as autoencoders — further enhance VHIs by capturing nonlinear dependencies and retaining critical health features through data compression and reconstruction. Recently, a novel skip-convolution generative adversarial network was proposed to construct HIs, demonstrating improved robustness and stability in tracking degradation trends (Qi et al., 2024).

The development of robust HIs, whether physical or virtual, is vital for accurate RUL prediction. Traditional PHIs remain integral to modern prognostics while evolving VHI techniques and advanced fusion methods continue to enhance predictive performance. Further details can be found in Guo et al. (2017), Liu and Chen (2019).

3.3. Health stage division

Dividing a system's lifecycle into distinct health stages is essential for accurate RUL prediction. Unlike fault diagnosis, which identifies fault patterns and severity at specific points, health stage division segments the continuous degradation process based on changes in health indicators over time. By categorizing degradation into stages, such as healthy, degraded, and critical, trends can be better modeled and analyzed (Hu et al., 2012).

Health stage division strategies for RUL prediction are broadly classified into two-stage and multi-stage approaches. The two-stage division is suitable when degradation trends in the unhealthy stage are consistent, allowing the use of a unified degradation model. For instance, a framework developed by Wang (2002) uses monitored condition data to predict RUL. A nonlinear Wiener process-based two-stage model proposed by Lin et al. (2021) captures complex degradation trajectories. More recently, an advanced two-stage method combining adversarial strategies and adaptive thresholds was introduced by Liu et al. (2024b), addressing data inconsistencies and improving prediction accuracy under variable conditions. Additionally, a two-stage approach incorporating both first predicting time detection and predictive uncertainty was proposed by Chen et al. (2023), demonstrating superiority in bearing applications.

Conversely, for systems with more complex degradation patterns due to varying fault modes or operational conditions, multi-stage division is required. This approach further segments the unhealthy stage into sub-stages to capture varying degradation trends. Battery health estimation was enhanced by Li et al. (2024a) through leveraging PCA and HIs to refine stage division. Meanwhile, clustering techniques such as K-means (Scanlon et al., 2012) and HMMs (Giantomassi et al., 2011) are frequently used to characterize intricate degradation trajectories.

In summary, health stage division plays a critical role in ensuring precise and adaptable predictions, particularly for systems with non-uniform degradation patterns. Without appropriate stage division, predictions may lack the precision needed to match real-world machinery behavior. Further details can be found in Liu et al. (2024b).

3.4. RUL prediction

RUL refers to the estimated time before a machine reaches a failure threshold, beyond which its performance becomes unacceptable. Following feature extraction and data preprocessing, the RUL prediction process begins with selecting an appropriate model, depending on the data characteristics and application requirements. The selected model is then trained using historical or simulated data, with parameters fine tuned to accurately reflect degradation patterns.

Once trained, the model is applied to realtime or incoming sensor data to generate RUL estimates. These predictions are dynamic and continuously updated as new data become available, allowing the model to adapt to changing operational conditions. At this stage, handling uncertainty becomes crucial. Techniques such as Monte Carlo

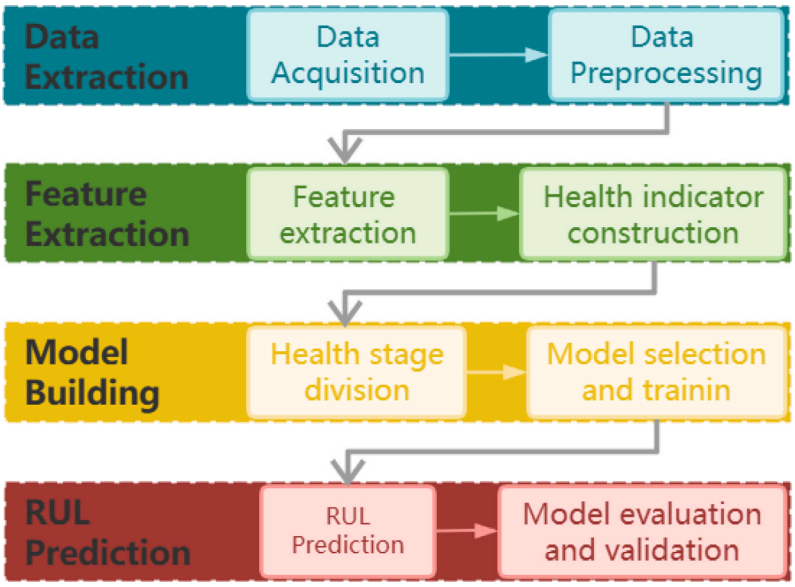


Fig. 5. RUL Prediction Process.

simulations and Bayesian inference are often employed to provide probabilistic predictions or confidence intervals, thereby accounting for variability in system degradation behavior.

After generating RUL predictions, model accuracy is evaluated using performance metrics such as Mean Squared Error (MSE) or Root Mean Squared Error (RMSE). Validation, through cross-validation or testing on independent datasets, is essential to ensure reliability across diverse operating conditions. This step is critical to confirm that the model remains robust when deployed in realworld scenarios. For a comprehensive review of evaluation metrics, refer to Lei et al. (2018).

In summary, RUL prediction involves model selection and training, real-time deployment with continuous updates, and rigorous performance evaluation. Incorporating uncertainty quantification and validation strategies not only improves prediction accuracy but also enhances the model's applicability in industrial settings, supporting informed maintenance planning and operational decision-making. Further details can be found in Ge et al. (2025), Zeng et al. (2024)

4. RUL prediction datasets

In the field of RUL prediction, access to high-quality datasets is crucial for the development, validation, and benchmarking of predictive models. Such datasets enable models to learn from realistic operational scenarios and improve their ability to generalize across different mechanical systems and working conditions. Commonly used datasets in this field originate from a range of applications, including aerospace engines, bearing systems, and lithium-ion batteries. In the following section, we briefly introduce the most widely adopted datasets, organized by application area, and highlight their relevance to RUL prediction tasks.

4.1. Aerospace and turbine engines

In the aerospace domain — particularly for turbomachinery — RUL prediction relies on both data-driven and physics-based methods to assess component degradation under complex operational conditions. Among the publicly available datasets, NASA's Turbofan Engine Degradation Simulation Dataset (Saxena & Goebel, 2008), developed using the C-MAPSS model, is the most widely used benchmark in this field. The dataset provides realistic simulations of turbofan engine degradation, establishing itself as a standard reference for developing and validating prognostic models.

The C-MAPSS dataset comprises four subsets (FD001 to FD004), each representing distinct operating conditions and fault modes across multiple engines. It includes run-to-failure time series data with sensor readings such as temperature, pressure, and rotational speed. Both training and testing sets are available, allowing researchers to evaluate model performance under variable operational scenarios. A key advantage of this dataset is its simulation of fluctuating operational conditions, which introduces diversity in degradation patterns. This makes the dataset especially valuable for studying how environmental and usage variability affect system health over time. Furthermore, the large volume of run-to-failure samples offers a rich training ground for data-driven models that require extensive input to generalize effectively.

Owing to its comprehensiveness and high-resolution multi-sensor data, the C-MAPSS dataset has been extensively used in RUL prediction research. It supports a wide range of techniques, including CNNs, LSTM networks, data fusion methods, unit-to-unit similarity modeling, and health stage assessment. The continued use of this dataset highlights its importance as a foundational resource for benchmarking advanced prognostic algorithms in aerospace applications. Further details about this dataset and associated research can be found in Vollert and Theissler (2021).

4.2. Bearing systems

Bearings are vital components in industrial machinery, and their failure can cause significant downtime and costly repairs. Two widely referenced datasets for bearing health monitoring and RUL prediction are the FEMTO Bearing Dataset (Nectoux et al., 2012) and the IMS Bearing Dataset (Qiu et al., 2006), both extensively used for evaluating predictive model performance.

- FEMTO Bearing Dataset: Published by the FEMTO-ST Institute and introduced at the 2012 IEEE PHM conference, this dataset was collected from the PRONOSTIA platform, designed for accelerated degradation testing of rolling bearings. A high radial force exceeding the maximum dynamic load was applied to simulate wear while maintaining a stable rotational speed. Vibration signals and temperature were captured using two accelerometers and a thermocouple, with failure defined as vibration amplitudes exceeding 20 g. This dataset has been widely used to develop RUL prediction models, focusing on time-domain and frequency-domain feature extraction for fault detection. This dataset serves

as a key resource for vibration-based health monitoring, enabling real-time tracking of bearing degradation under stress. Further details can be found in Lin et al. (2024), Nectoux et al. (2012).

- IMS Bearing Dataset: Provided by the University of Cincinnati's Center for Intelligent Maintenance Systems and shared via NASA's Prognostic Data Repository, this dataset comprises three subsets of rolling bearing degradation tests. Four Rexnord ZA-2115 double-row bearings were tested with accelerometers mounted on housings to capture vibration data under varying load conditions. An oil circulation system ensured lubrication, while a magnetic plug collected debris in the feedback pipe, signaling degradation. Tests were stopped when debris accumulation exceeded a threshold, with fault patterns recorded for each bearing. High-resolution vibration signals (20.48 kHz sampling rate) enable detailed frequency analysis, facilitating the extraction of frequency-domain features and monitoring the degradation of components like rollers and races. Further details can be found in Qiu et al. (2006), Yan et al. (2020).

4.3. Battery systems

Accurate RUL prediction is essential for enhancing the reliability, safety, and longevity of battery systems, which are crucial to the efficiency of modern technologies. Two benchmark datasets for battery health estimation and RUL prediction are the NASA Battery Prognostics Dataset (Saha & Goebel, 2007) and the CALCE Battery Data Set (Center for Advanced Life Cycle Engineering (CALCE), 2025).

- NASA Battery Prognostics Dataset: Widely recognized in the Battery Health Management (BHM) community, this benchmark records lithium-ion cells operating under varied charge–discharge regimes. It tracks voltage, temperature, and current in high resolution, thereby supporting both physics-based degradation modeling and data-driven learning. Classical methods such as SVM and GPR exploit these measurements to estimate degradation trajectories and RUL; meanwhile, the rich time-series structure enables LSTM and other deep networks to capture nonlinear ageing effects. Owing to its breadth of operating conditions, the dataset remains indispensable for testing model robustness and generalization. Further details can be found in Hasib et al. (2021), Saha et al. (2008).
- CALCE Battery DataSet: Compiled by the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland, this dataset monitors lithium-ion batteries exposed to diverse stress factors, including variable discharge rates and ambient temperatures. Its chief value lies in real-world relevance, providing long-term capacity, internal resistance, and voltage-drop profiles—metrics that are critical for electric vehicle and aerospace applications. Researchers have leveraged these data to build models spanning simple exponential fits to advanced RNNs and PFs. Moreover, CALCE dataset frequently underpin hybrid schemes that fuse physics-based insights with machine learning algorithms, thereby boosting RUL-prediction accuracy. Some example applications can be found in Feng and Xue (2025), Zheng et al. (2016).

5. RUL prediction tools

The practical implementation of RUL prediction systems primarily relies on the Python programming language. Python has become the standard in this field due to its rich set of libraries and strong community support. Its ecosystem provides general-purpose tools for data handling and machine learning, as well as domain-specific libraries for prognostics tasks.

5.1. Data processing libraries

Python libraries such as **NumPy** and **Pandas** are widely used for numerical computation and data manipulation. **NumPy** supports efficient multi-dimensional array operations, while **Pandas** offers structured data handling through DataFrames. For preprocessing, **Scikit-learn** provides pipelines that integrate normalization methods with traditional machine learning algorithms. These tools help ensure consistency and reproducibility in the RUL modeling workflow.

5.2. Deep learning frameworks

Deep learning models are often used in modern RUL studies. The two dominant frameworks are **TensorFlow** (with the high-level **Keras** API) and **PyTorch**. **TensorFlow**'s **tf.data** module efficiently handles large-scale time-series data with sliding-window sampling and batching. **PyTorch**, known for its dynamic computation graph, is favored for its flexibility during model development and debugging. Both frameworks are well-supported and have extensive model libraries for sequence modeling.

5.3. Domain-specific libraries

Several Python libraries support feature extraction and reliability modeling in RUL tasks. Table 3 summarizes the key domain-specific libraries used in the literature.

These libraries help accelerate development by providing tested routines for RUL-related data preparation and modeling. Combining general and domain-specific tools allows for efficient and reproducible implementation of RUL systems.

6. Scientometric analysis for research atricle

6.1. Data source and methods for literature analysis

The WoS is a premier global citation database covering a broad range of disciplines including the hard sciences, social sciences, arts, and humanities. In this study, we selected the WoS Core Collection as our primary data source to ensure high academic quality and indexing consistency. The search query "Remaining useful life prediction" was applied across the title, abstract, author keywords, and Keywords Plus fields, generating an initial dataset of 3796 records. To ensure relevance to the engineering-focused RUL research field, a multi-stage filtering process was conducted. First, only English-language, peer-reviewed journal articles were retained, thereby excluding conference proceedings, book chapters, and preprints. Second, titles and abstracts were manually screened to eliminate documents unrelated to engineering prognostics and health management—such as those focused on biological, clinical, or other non-industrial contexts. This filtering process yielded a final dataset of 3442 qualified articles. The dataset was then exported in plain text format for subsequent scientometric analysis and knowledge visualization using CiteSpace.

The number of published documents is a crucial indicator of research activity in a field (Yuan & Sun, 2022). As illustrated in Fig. 6, research on RUL prediction in WoS has grown notably over the years. From 1997 to 2011, the volume of articles in the databases was low. However, from 2011 onward, WoS has seen rapid growth, peaking at 686 articles in 2023. This demonstrates a significant increase in international interest in RUL prediction research. The lower number of articles in 2024 is not due to a decline in interest but because the data collection was cut off in mid-year.

This study was mainly based on CiteSpace, a robust tool for scientometric analysis and information visualization, to investigate the research landscape for RUL prediction. Developed by Professor Chen, CiteSpace allows the identification of key research trends, collaboration

Table 3
Common domain-specific libraries used in RUL implementation.

Library	Functionality	Application area
tsfresh	Extracts a wide range of time-domain and frequency-domain features, including mean, variance, zero-crossing rate, FFT energy, and autocorrelation metrics.	Automated feature engineering for machine learning-based RUL regression and classification tasks, especially when domain knowledge is limited.
PyWavelets	Implements discrete wavelet transforms (DWT) for multi-resolution time-frequency analysis of non-stationary signals.	Useful for vibration-based diagnostics, anomaly detection, and transient signal decomposition in rotating machinery and complex equipment.
lifelines	Provides survival analysis models such as the Kaplan–Meier estimator and Cox proportional hazards model, supporting censored data and covariate effects.	Applied in reliability analysis and failure time modeling where the prediction is probabilistic and data may involve partial observations.

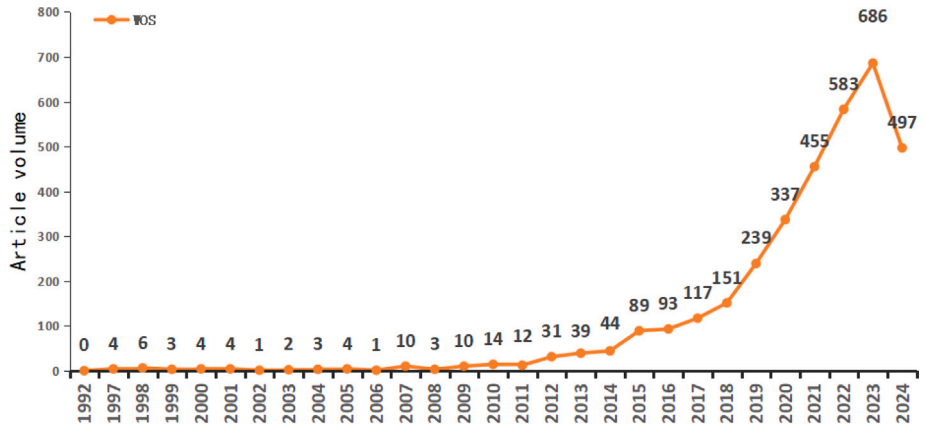


Fig. 6. Annual Volume of the RUL Research from the WoS Database.

networks, and emerging topics by analyzing citation data and visualizing relationships within the scholarly literature (Chen, 2004, 2006; Chen et al., 2016). This study uses CiteSpace version 6.3.R1. For the plain text files exported from WoS, the time slice was set to one year. Further more, We selected the node types 'Countries', and 'Institutions' for network visualization analysis, and 'Keywords' for co-occurrence analysis, clustering analysis, and burst detection analysis.

6.2. Overview of research areas

The study of RUL prediction spans diverse research areas, as shown in Fig. 7. Engineering dominates with 2477 articles, reflecting its central role in maintaining mechanical systems across aerospace, automotive, and manufacturing industries. Computer science, contributing 766 articles, enhances predictive accuracy through machine learning and data mining. Instruments and instrumentation, with 592 articles, focus on precise measurement and real-time data collection for accurate predictions. Fields like energy, operations research, automation, chemistry, materials science, physics, and telecommunications also provide valuable insights into system reliability. This interdisciplinary collaboration drives advancements in RUL models and improves the reliability and efficiency of various infrastructures.

6.3. Main research forces analysis

6.3.1. National collaboration analysis

A scientometric analysis conducted on 3442 WoS articles reveals the global distribution of RUL prediction research. Node size, connections, and centrality values (highlighted in purple for values >0.1) indicate article counts and collaborative influence among countries (Ma et al., 2023). The cooperation network parameters (N = 79, E = 263, Density = 0.0854) reflect a closely connected research community.

Table 4
Top 10 countries in RUL prediction research from the WoS database.

Rank	Country	Centrality	Count
1	China	0.25	2361
2	USA	0.27	479
3	France	0.17	192
4	England	0.28	157
5	Canada	0.03	149
6	South Korea	0.06	130
7	Italy	0.11	115
8	India	0.03	87
9	Singapore	0.00	79
10	Germany	0.07	61

Fig. 8 and Table 4 show China leading with 2361 articles, followed by the USA with 479 articles, which also exhibits high centrality, signifying broad influence. France and England, with 192 and 157 articles, play significant roles due to their strong collaborative networks. Other contributors, including Canada, South Korea, Italy, India, Singapore, and Germany, offer diverse insights, emphasizing the value of international cooperation in advancing predictive maintenance technologies.

6.3.2. Institutional collaboration analysis

Fig. 9 shows the academic collaboration of different institutions in RUL prediction field research, while Table 5 shows a detailed list of the volume of articles of the top 10 institutions.

From Fig. 9, we can conclude that the collaboration network in RUL prediction research highlights key institutions such as Beihang University, Xi'an Jiaotong University, Chongqing University, Chinese Academy of Sciences and Tsinghua University, which exhibit significant influence through extensive connections. Chinese dominance in this field is driven by substantial government funding, rapid industrial growth, and strong academic programs. Research clusters are evident

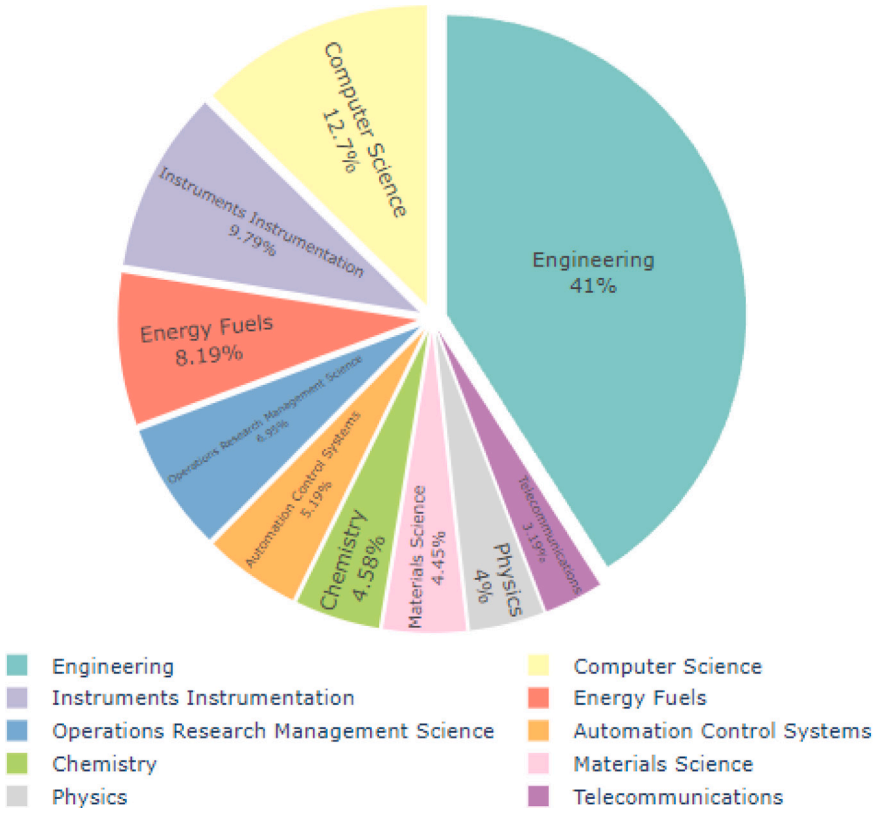


Fig. 7. Research Area Distribution for RUL-Related Studies.

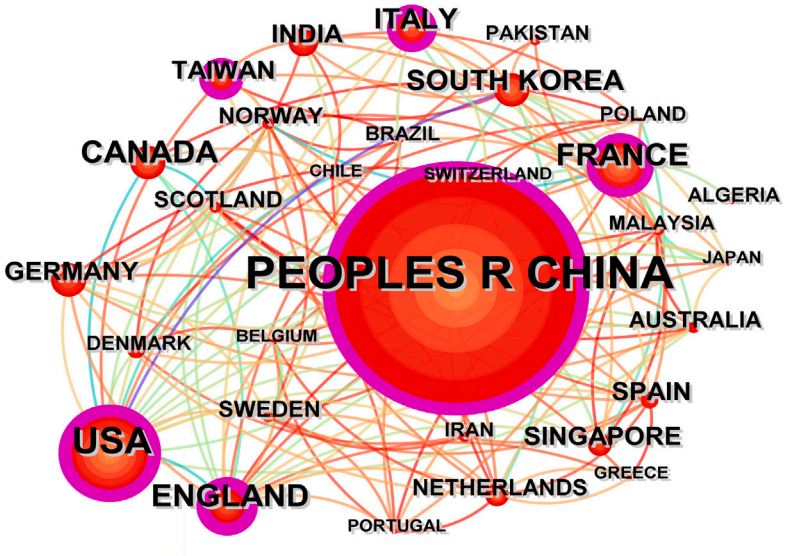


Fig. 8. National Cooperative Relationship Network.

among these institutions, such as those between Chongqing University and Shanghai Jiao Tong University, Xi'an Jiaotong University, Peking University, and Nanjing University of Aeronautics & Astronautics and the Chinese Academy of Sciences. European institutions like CNRS and the Polytechnic University of Milan contribute significantly due to established research traditions, collaborative frameworks such as Horizon 2020, and interdisciplinary approaches. North American institutions, such as Northeastern University and the University System of Ohio, benefit from a culture of technological innovation and cross-disciplinary research supported by federal grants and industry partnerships. To enhance future collaboration, promoting international partnerships, fostering cross-disciplinary research, and providing targeted policy and funding support are essential. These measures will drive greater innovation and impact in RUL prediction research, ensuring comprehensive and robust results.

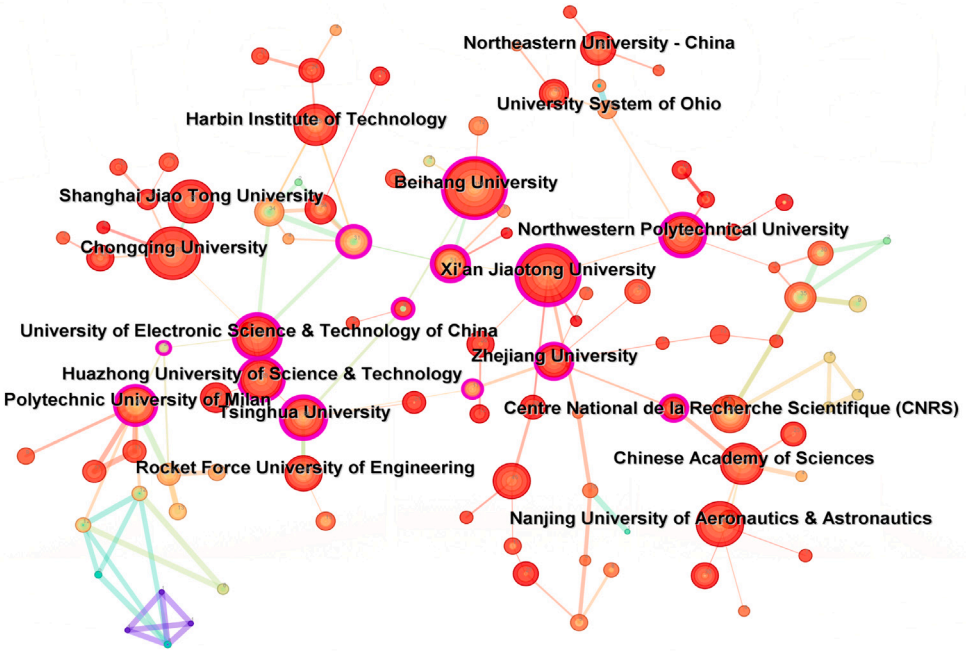


Fig. 9. Institution Cooperative Relationship Network.

Table 5
Top 10 institutions in RUL research from the WoS database.

Rank	Institution name	Record count
1	Beihang University	150
2	Xi'an Jiaotong University	124
3	Chongqing University	105
4	Nanjing University of Aeronautics & Astronautics	95
5	Shanghai Jiao Tong University	85
6	Chinese Academy of Sciences	75
7	Harbin Institute of Technology	75
8	University of Electronic Science & Technology of China	73
9	Centre National de la Recherche Scientifique	70
10	Rocket Force University of Engineering	65

6.4. Paper source analysis

Analyzing the articles in the WoS database provides valuable insights into the dissemination of research within the field of prediction. Understanding which journals are chosen most frequently by researchers helps identify the leading platforms for RUL-related articles, ensuring that research reaches the appropriate academic audience and garners the visibility it deserves. To this end, we have compiled and listed the top 20 journals ranked by the number of articles, as shown in Table 6.

6.5. Research themes and hotpots evolution analysis

Temporal clustering analysis of keywords offers valuable insights into the evolution of research themes and trajectories in RUL prediction. We employed CiteSpace to conduct keyword co-occurrence and clustering analysis. The time slicing was configured from 1997 to 2024 with one-year intervals. "Keyword" was selected as the node type, and the g-index ($k = 5$) was used as the selection criterion. To preserve essential structural paths, the Pathfinder algorithm was applied for network pruning. Cluster labeling was performed using the Log-Likelihood Ratio (LLR) method. The resulting network consisted of 263 nodes and demonstrated strong clustering performance, with a modularity Q of 0.8692 and a mean silhouette score of 0.9603. These values indicate a well-structured network with high internal consistency among clusters. The keyword clustering timeline is illustrated in Fig. 10.

The timeline categorizes research evolution into three main themes: Common Research Methods, Prevalent Research Issues, and Typical Research Objects. Keywords are grouped into 13 categories, highlighting distinct trends and shifts in focus over time, providing a clear view of the progression of the field.

6.5.1. Common research methods

The evolution of RUL prediction methods is captured in clusters 0 (remaining useful life), 2 (condition monitoring), 3 (feature extraction), 4 (regression analysis), 8 (machine learning), 10 (particle filter), 11 (unscented particle filter) and 12 (bayesian inference).

During the early phase (1997–2010), 'regression analysis' emerged as a key method, leveraging stochastic processes and probabilistic models to analyze equipment state changes and faults. Technologies such as condition monitoring, which used sensor data to assess equipment health, laid the foundation for later data-driven approaches (Chelidze & Cusumano, 2004; Goode et al., 2000).

From 2010 to 2019, the focus shifted to data-driven methods, driven by advancements in computational power. Probabilistic models declined in favor of techniques that handle high-dimensional data. Keywords like 'particle filter' (Orchard & Vachtsevanos, 2009; Saha et al., 2009, 2008), 'support vector machine' (Lu & Wang, 2020; Xue et al., 2020), 'condition-based maintenance' (Zhou et al., 2011) and 'feature extraction' (Mosallam et al., 2013; Scanlon et al., 2012) became prominent, marking significant progress in machine learning applications for RUL prediction.

With enhanced computational power and data processing capabilities, deep learning techniques like 'convolutional neural networks' and 'long short-term memory' networks have become key in RUL prediction research (An et al., 2020; Liu et al., 2019b). Initially popular in image processing and time series analysis, these methods now excel in predicting equipment failures, showcasing the growing complexity and accuracy of modern predictive models.

Since 2019, advanced techniques like 'attention mechanism' (Zhang et al., 2020b) and 'transfer learning' (Gribbestad et al., 2021; Zeng et al., 2021) have emerged to improve prediction accuracy and robustness. These methods focus on critical features and adapt knowledge

Table 6
Top 20 source of papers in the RUL research from the WoS database.

Rank	Publication titles	Record count	% Share
1	Reliability Engineering and System Safety	243	6.76
2	IEEE Access	154	4.28
3	Mechanical Systems and Signal Processing	125	3.48
4	IEEE Transactions on Instrumentation and Measurement	123	3.42
5	Sensors	100	2.78
6	Measurement	94	2.61
7	IEEE Transactions on Reliability	87	2.42
7	Measurement Science and Technology	87	2.42
9	Energies	83	2.31
10	Journal of Energy Storage	81	2.25
11	Applied Sciences Basel	79	2.20
12	IEEE Transactions on Industrial Informatics	60	1.67
13	IEEE Transactions on Industrial Electronics	59	1.64
14	Applied Energy	54	1.50
14	Energy	54	1.50
16	Microelectronics Reliability	53	1.47
17	Quality and Reliability Engineering International	46	1.28
18	Journal of Power Sources	44	1.22
19	IEEE Sensors Journal	41	1.14
20	Applied Soft Computing	36	1.00
20	Engineering Applications of Artificial Intelligence	36	1.00

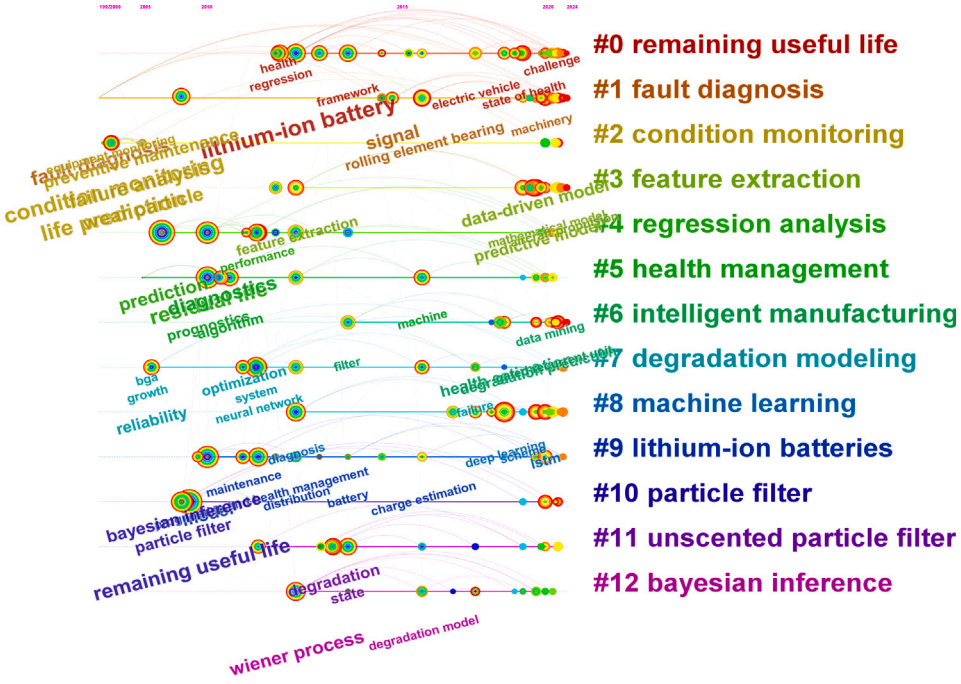


Fig. 10. Keyword Timeline in the RUL Research (1997–2024) from the WoS Database.

across domains, reflecting the trend toward more intelligent systems capable of handling complex datasets.

6.5.2. Prevalent research issues

Clusters 1 (fault diagnosis), 5 (health management), and 7 (degradation modeling) outline the evolving research challenges in RUL prediction.

In the early stage, before 2010, the focus was primarily on health management and predictive maintenance. 'fault diagnosis' emerged as a prominent keyword, reflecting the efforts to identify early signs of equipment failure and improve maintenance strategies.

Between 2010 and 2015, research began to address the increasing complexity and uncertainty of RUL prediction models. The emergence of keywords such as 'Bayesian inference' indicated a growing interest in probabilistic reasoning to enhance model reliability (Pasquier & Smith, 2015). Simultaneously, terms like 'reliability' and 'management' pointed toward the development of systematic frameworks for equipment operation and decision-making (Ramasso & Gouriveau, 2014).

From 2015 to 2020, the focus shifted to precise health estimation, with keywords such as 'health estimation' and 'health indicator' reflecting advances in feature engineering and analytics for equipment condition assessment (Dolenc et al., 2017).

Since 2020, research has increasingly incorporated advanced analytics and intelligent technologies. Topics such as 'battery degradation' and 'performance degradation', particularly driven by the widespread application of lithium-ion batteries, have gained prominence. These trends highlight ongoing challenges in maintaining system reliability under diverse and evolving operational conditions.

6.5.3. Typical research objects

Clusters 6 (intelligent manufacturing) and 9 (lithium-ion batteries) highlight key research subjects in RUL prediction, including 'rolling bearing', 'lithium-ion batteries', 'electric vehicles', and 'state of charge'.

Each object requires tailored methods to address its unique characteristics, reflecting the diverse applications of RUL research.

In the field of lithium-ion batteries, researchers focus on modeling electrochemical degradation mechanisms and estimating key states such as 'state of charge' and 'state of health'. These efforts aim to improve the reliability and safety of battery-powered systems, particularly in electric vehicles and energy storage applications. Techniques such as data-driven modeling, hybrid frameworks, and advanced state estimation algorithms have been widely adopted to capture complex aging patterns and optimize usage strategies.

Meanwhile, the emergence of intelligent manufacturing as a research cluster indicates a broader shift toward system-level prognostics in Industry 4.0 environments. In these contexts, RUL prediction supports smart factories through real-time monitoring, cross-device data fusion, and predictive analytics. The integration of machine learning and IoT infrastructure enables proactive maintenance strategies across production lines, improving equipment availability and operational efficiency.

These research directions demonstrate how RUL methodologies are increasingly tailored to specific technological contexts, ranging from component-level systems to cyber–physical manufacturing environments.

6.6. Keywords brutness analysis

Emerging keywords in RUL prediction highlight recent and future research trends, with keyword burstiness signaling areas of rapid advancement (Zhang et al., 2020a). As shown in Table 7, keywords such as 'condition monitoring' (2000–2018) reflect the transition from reactive to predictive maintenance, enabling real-time health assessments and operational efficiency improvements.

From 2009 to 2019, the rise of 'particle filter' underscores the adoption of probabilistic and model-based methods for precise state estimation in complex systems. Between 2013 and 2020, keywords like 'support vector machines' and 'degradation modeling' indicate advances in understanding equipment degradation, addressing challenges in dynamic environments.

The burst of deep 'learning' (2016–2019) marks the impact of advanced neural networks for processing large datasets, fine-tuning predictive maintenance strategies, and enhancing system adaptability. Recent trends, such as 'recurrent neural networks' and 'temporal convolutional networks' (2022 onward), focus on time-series prediction challenges, critical for real-time RUL estimation.

Keywords like 'battery management systems' and 'wind turbines' highlight growing interest in improving the performance and lifespan of key components, particularly in energy storage and renewable energy sectors.

7. Trends and future perspectives

Based on the findings and conclusions in Section 6, combined with the latest literature from the past two years, this analysis provides critical insights into RUL prediction research trends, technologies, and future directions.

7.1. Emerging trends and hotspots

Keyword analysis reveals clear developmental trends and technological progression in RUL prediction research, which not only reflects the evolution of methodological approaches but also underscores the growing intersection with practical industrial demands.

First, there is a strong shift toward leveraging advanced machine learning and deep learning techniques, particularly RNNs, LSTM, and TCNs. These methods are increasingly favored because they can model sequential dependencies in time-series degradation data. At the same time, advanced statistical tools like Bayesian inference and GPR have been adopted to improve uncertainty estimation, which is very important in safety-sensitive applications. Second, the growing focus on sustainable development has made lithium-ion batteries management a key area, especially for electric vehicles and renewable energy systems. Real-time health estimation and predictive maintenance are becoming more important as core industrial needs. What is more, integrated systems that combine condition monitoring, intelligent data analytics, and online decision-making are gradually becoming the norm in modern industrial settings.

7.2. Implications for future research

7.2.1. Multi-source data and feature fusion

Effectively leveraging multi-source data fusion is crucial for improving the accuracy and robustness of RUL prediction models. Integrating various signals to comprehensively capture the degradation patterns of equipment remains a key challenge. For instance, Chen et al. (2024) proposed an RUL prediction method based on the deep fusion of nonlinear features from multi-source data, while Wei et al. (2024) demonstrated the potential of multi-source information fusion using high-dimensional kernel density estimation with an adaptive relative density window width. Recent studies further highlight the importance of spatiotemporal information integration. For instance, Li et al. (2023a) introduced the STAIRnet method, which combines a spatiotemporal attention module and an information reinforcement mechanism to adaptively encode signals in temporal and spatial dimensions, further enhancing feature extraction capabilities and prediction accuracy. Similarly, Yao et al. (2024) presentd the multi-scale PatchTCN-Mixer model, which significantly improves RUL prediction accuracy and generalization by leveraging subseries segmentation and spatiotemporal semantic extraction. These advancements underline the importance of multi-source data fusion and advanced feature extraction techniques in building more robust RUL prediction frameworks.

7.2.2. Uncertainty quantification

Addressing prediction uncertainty is critical for ensuring reliability, especially in safety-critical applications. The integration of methods to manage both epistemic and aleatoric uncertainties can significantly enhance the robustness of RUL prediction models. This approach enables more reliable predictions even under limited supervision or minimal labeled data (Pan et al., 2024).

For example, Zeng et al. (2024) proposed a self-driven RUL prediction method named Bayesian domain-adversarial regression adaptation, which not only adapts across domains but also provides credible intervals for the prediction results. Ge et al. (2025) developed a modernized temporal convolutional network with embedded uncertainty quantification for high-accuracy bearing RUL prediction.

In addition, local–global modeling paradigms, as explored in uncertainty-aware device-free localization tasks (Zhang et al., 2024), demonstrate that introducing clustering-based variance constraints can further improve robustness under complex, uncertain environments. Similar mechanisms could be valuable in RUL contexts, particularly when dealing with heterogeneous operational data and non-stationary degradation patterns.

7.2.3. Scalability and flexibility

Developing scalable and adaptable models is crucial for application across diverse industries. Future solutions should focus on creating industry-specific models that retain knowledge across learning phases, adapt to dynamic environments, and reduce dependence on extensive datasets, ensuring efficiency without compromising accuracy. In this context, continual learning has emerged as a promising approach, enabling deep prognostics networks to incrementally learn from new data without the need for exhaustive retraining. For example, recent work has proposed a complementary continual learning framework

Table 7
Analysis of keywords burstiness from the WoS database.

Keywords	Strength	Start	End	1997 - 2024
condition monitoring	4.27	2000	2018	
condition-based maintenance	3.86	2000	2017	
particle filter	27.76	2009	2019	
bayesian inference	10.33	2009	2017	
support vector machine	4.93	2013	2020	
wiener process	4.90	2014	2018	
remaining useful life	3.71	2009	2017	
data-driven prognostics	3.09	2014	2019	
rolling element bearing	3.24	2015	2016	
degradation modeling	8.21	2016	2019	
remaining life assessment	3.92	2017	2019	
gaussian process regression	3.62	2017	2020	
data-driven model	3.36	2018	2020	
wind turbine	3.24	2018	2019	
recurrent neural network	6.80	2019	2021	
artificial neural network	5.77	2019	2021	
battery management system	3.27	2019	2020	
genetic algorithm	3.09	2019	2022	
adaptation model	4.13	2020	2024	
maintenance engineering	3.22	2020	2022	
logic gate	5.37	2021	2022	
task analysis	3.51	2021	2022	
market research	3.82	2022	2024	
temporal convolutional network	3.32	2022	2024	

incorporating long-term potentiation and associative replay mechanisms to address challenges such as knowledge retention and dynamic adaptation during learning stages (Ren et al., 2024). Furthermore, challenges such as insufficient training data and model efficiency have also been tackled using techniques like transfer learning. For instance, a transfer learning-based training method coupled with a distillation pruning algorithm has been shown to effectively reduce the demand for training data while significantly minimizing model parameters, without sacrificing prediction performance (Zheng et al., 2024).

Recent advances in online adaptive modeling have significantly enhanced real-time RUL prediction capabilities. For example, Wang et al. (2025a) proposed an online adaptive multi-degradation model that introduces two key innovations: a dynamic sliding window strategy for automatic first prediction time determination, replacing subjective threshold-based approaches, and an online parallel prediction algorithm that continuously evaluates and selects the optimal degradation model from multiple candidates.

Notably, an adversarial contrastive learning approach proposed by Shang et al. (2025) addresses single-source generalization challenges through pseudo-domain generation and domain-invariant feature extraction. These developments align with broader progress in adaptive learning systems, particularly online learning with domain adaptation (Zhang et al., 2025), presenting a promising direction for next-generation prognostic systems facing dynamic operational conditions.

7.2.4. Technological integration

The integration of IoT, big data, and machine learning presents new opportunities for RUL prediction. Advances in AI-based methods for monitoring and fault prediction, as well as the emergence of large-scale AI models like ChatGPT and DALL-E, could transform predictive maintenance, although further exploration is needed to align these tools with PHM applications (Li et al., 2023b). Additionally, the incorporation of attention mechanisms offers new possibilities for RUL prediction by enhancing models' ability to focus on critical features, improving accuracy and interpretability. For instance, in You et al. (2023), a 3D Attention-enhanced hybrid neural network was introduced to address the RUL prediction challenges of turbofan engines. Future research could explore optimizing attention-based models to address challenges such as uneven data distribution and dynamic feature importance, further advancing intelligent PHM solutions.

7.3. Summary

The integration of IoT, cloud computing, and AI has unlocked new opportunities for predictive maintenance. Specifically, AI-driven sensor networks now enable autonomous data acquisition and real-time fault diagnosis. At the same time, the rise of large language models and generative AI has highlighted their potential for PHM tasks (Li et al., 2023b). In addition, digital twin frameworks are being widely adopted to simulate physical asset behavior, allowing real-time monitoring, scenario testing, and predictive analysis in virtual environments. Another key trend is the emergence of explainable AI as a critical research frontier, especially in regulated or safety-critical domains. Furthermore, by providing transparent and interpretable model outputs, explainable AI enhances trust and promotes broader adoption of RUL prediction systems.

The ongoing evolution of RUL prediction is fueled by the convergence of emerging technologies and interdisciplinary collaboration. These advancements hold significant promise for improving equipment reliability and operational resilience across industries.

In summary, this study offers a comprehensive overview of the RUL prediction landscape — encompassing model-based, data-driven, and hybrid methods — along with representative datasets, tools, and research trends. By combining a methodological review with scientometric analysis, it provides both technical insights and strategic guidance. Moving forward, sustained focus on innovation, integration, and practical implementation will be essential for advancing reliable and intelligent prognostic systems.

CRediT authorship contribution statement

Yitong Liu: Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Jiarui Wen:** Methodology, Software, Validation, Writing – review & editing. **Guoqiang Wang:** Conceptualization, Methodology, Funding acquisition, Supervision, Writing –review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author would like to thank the reviewers for their insightful comments, and this work is supported by the National Natural Science Foundation of China (Nos. 12171307, 11971302).

Data availability

Data will be made available on request.

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